TITLE: Centaur-Chip: A Topological Photonic Accumulator Swarm for Deceleration and Data Return from Alpha Centauri

AUTHOR: Ruslan A. Kudinov

AFFILIATION: Independent Researcher, Russian Federation

EMAIL: Saint_Nalsur@mail.ru

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ABSTRACT

The interstellar lightsail concept (Breakthrough Starshot) faces a fundamental deadlock: the absence of a braking mechanism at the target system and the diffraction limit of communication for a single gram-scale microprobe. This paper proposes the "Cyclone-M" class architecture — a swarm of nanostructured "Centaur-Chip" probes — in which these limitations are resolved through three interconnected physical solutions: 1) A hollow-core photonic crystal ring with topological protection (Möbius-strip geometry) for photon storage and radiation pressure compensation; 2) A dual-wavelength Doppler laser cooling method enabling cryogenic thermal management of the probe directly within the acceleration beam; 3) Sub-aperture swarm synchronization via mutual quantum-optical correlation into a phased array antenna for coherent data transmission to Earth via the Solar Gravity Lens (SGL). An energy density analysis shows the accumulated field remains 4 orders of magnitude below the Schwinger limit, though requiring ultra-high vacuum cavity preparation. A sacrificial layer erosion estimate confirms that material loss from relativistic interstellar gas impact is manageable ($\sim 1 - 10 \mu \mathrm{m}$). The proposed architecture enables a full mission cycle: acceleration, active braking via an "energy ram," and scientific data return without physical carrier recovery.

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1. INTRODUCTION: BOTTLENECKS OF THE CLASSICAL SAIL SCHEME

The straight flyby of gram-scale probes through the Alpha Centauri system at 0.2c, proposed by the Starshot initiative, yields only seconds of observation. Two engineering chasms prevent the transformation of a flyby into a full-fledged mission:

1. The Braking Paradox: Flipping the spacecraft for retrograde thrust at relativistic speeds is impossible due to the enormous dynamic pressure of the interstellar medium.
2. The Energy-Information Collapse: A single chip cannot focus a signal across 4.3 light-years due to the fundamental diffraction limit.

2. ARCHITECTURE: HOLLOW-CORE TOPOLOGICAL PHOTONIC ACCUMULATOR

2.1 Hollow-Core Photonic Crystal Ring with Topological Protection

Unlike classical optical resonators where photons interact with matter and lose energy to heat, our design employs a hollow-core photonic crystal waveguide bent into a closed ring. Light circulates in the evacuated core channel, while the topological band gap is realized in the periodic cladding structure. The electromagnetic field penetrates the walls only at the evanescent length, excluding dissipative heating of the bulk material. The interaction is field-structure, not field-atom — enabling contactless photon storage without conventional mirrors.

2.2 Radiation Pressure Compensation (Mobius Strip Geometry)

The primary threat during gigawatt-level light pulse accumulation is radiation pressure tearing the structure apart. To solve this, the closed ring waveguide is geometrically shaped as a Mobius strip. Light completing a full cycle traverses the path with inverted polarity twice. The pressure vectors of photons mutually compensate on opposing walls, converting destructive force into zero net impulse relative to the crystal lattice of the hull.

2.3 Charging Strategy: In-Situ Stellar Refueling

The accumulator does not store energy for the entire interstellar cruise. The primary charging cycle occurs upon close approach to the target star (e.g., Proxima Centauri). The nanorings switch to absorption mode, harvesting stellar radiation in real time. The accumulated energy is then used immediately for the braking maneuver — eliminating the requirement for prohibitively high energy density during transit.

2.4. Adiabatic Mode Transformation and Topological Phase Shift

To rigorously demonstrate the absence of parasitic scattering and reflection at the geometric closure of the waveguide, we analyze the propagation within the framework of topological photonics. The hollow-core photonic crystal lattice exhibits a synthetic gauge field characterized by spin-momentum locking, where the pseudo-spin state $|\sigma\rangle$ of the electromagnetic mode is strictly bound to its wavevector $\mathbf{k}$. The Möbius strip configuration introduces a slowly varying metric perturbation $g_{\mu\nu}(\theta)$, where $\theta \in [0, 2\pi]$ is the azimuthal coordinate along the ring. Since the pitch of the structural twist satisfies the adiabatic condition: $\frac{\lambda}{L_{twist}} \ll 1$ (where λ is the operational wavelength and $L_{twist} \approx 3$ m is the total ring perimeter), the system obeys the adiabatic theorem. The propagating mode undergoes an uninterrupted evolution, preventing inter-mode coupling into radiative or backward-scattered continuum states. Upon completing a full spatial revolution ($\theta = 2\pi$), the boundary condition dictates an inversion of the polarization vector, mapping $|\sigma\rangle \to -|\sigma\rangle$. This geometrical evolution accumulation is equivalent to a geometric phase shift (Berry phase) of exactly $\gamma = \pi$. Consequently, the destructive self-interference typically observed at the junction of non-topological phase-shifted resonators is structurally eliminated. The configuration ensures an adiabatic mode tracking during polarization inversion, suppressing standard bending and scattering losses inherent to conventional high-Q optical cavities.

3. ENERGY DENSITY AND SCHWINGER LIMIT ANALYSIS

A critical examination of the energy density within the accumulator is required. Assume the accumulator captures 10 GJ of stellar energy over 1 hour at the target star. For a ring with a mode volume of 1 mm^3 (length ~ 3 m, cross-section $\sim 0.3\times 0.3$ mm), the field energy density is: $\rho = 10^{10} \text{ J} / 10^{-9} \text{ m}^3 = 10^{19} \text{ J/m}^3$. This corresponds to an electric field strength of $\sim 10^{14} \text{ V/m}$. The Schwinger limit — where the vacuum becomes nonlinear and electron-positron pair production begins — is $\sim 10^{18} \text{ V/m}$. The operational field is therefore 4 orders of magnitude below the Schwinger limit, remaining within the domain of classical electrodynamics. However, at such field intensities, any micro-roughness or residual particle within the cavity will trigger avalanche breakdown. This imposes a strict engineering requirement: the cavity must undergo pre-charging laser ablation cleaning and be maintained at ultra-high vacuum (UHV) levels before energy accumulation begins. This is a severe technological challenge, but not a fundamental physical prohibition.

4. DUAL-VECTOR "ENERGY RAM" AND DECELERATION

Upon approach to the target at 0.5c - 0.7c, the accumulated light is redirected to the forward hemisphere and ejected in a narrow beam strictly along the trajectory. This method solves two problems simultaneously:

· Photonic Deceleration: Controlled thrust is generated, reducing velocity to orbital insertion values without spacecraft rotation.
· Plasma Wedge: The counter-directed coherent radiation vaporizes microparticles and interstellar gas ahead of the probe, reducing the density of impacting material by orders of magnitude.

Mechanical loading consideration: The ejection of stored photons creates a reaction force on the chip via the output aperture. To avoid destructive peak stresses, the accumulated energy is released quasi-statically over 10^4 - 10^5 seconds, maintaining power levels below the material damage threshold. Distributed multi-aperture emission further spreads the mechanical impulse across the chip's crystal lattice.

Caveat: At 0.5c, interstellar hydrogen atoms possess energies of tens of MeV. Even with laser ablation, residual relativistic particles and secondary radiation will impact the forward face. The "absolutely transparent zone" is an idealization. The chip's leading edge is doped for passive radiation hardness and operates in a sacrificial degradation mode, accepting a finite erosion rate over the mission duration.

Sacrificial Layer Erosion Estimate

During the braking phase, the probe decelerates from 0.5c to orbital velocity ($\sim 10^{-4}c$) over a duration $t_{\text{brake}} \sim 10^{5}$ s (approximately 1 day) while traversing a column density of interstellar gas. The interstellar hydrogen number density toward Proxima

Centauri is n \sim 0.1 \, \text{cm}^{-3}. At 0.5c, each proton impacts with kinetic energy E_p = (\gamma - 1) m_p c^2 \approx 0.15 m_p c^2 \approx 140 MeV. The effective fluence on the forward face (area A \sim 1 \, \text{cm}^2) is: F = n \cdot v \cdot t_{\text{brake}} \cdot A \approx (10^5 \, \text{m}^{-3}) \cdot (1.5 \times 10^8 \, \text{m/s}) \cdot (10^5 \, \text{s}) \cdot (10^{-4} \, \text{m}^2) \approx 1.5 \times 10^{14}. Assuming a diamond-like carbon sacrificial layer with density \rho = 3.5 \, \text{g/cm}^3 and sputter yield Y \sim 0.1 atoms per incident proton (at 140 MeV, electronic stopping dominates; nuclear spallation yield is low, but we adopt a conservative upper bound), the eroded volume per proton is Y / (\rho N_A) \sim 5 \times 10^{-25} \, \text{cm}^3. Total eroded thickness: \Delta x = \frac{F \cdot Y}{\rho N_A A} \approx \frac{(1.5 \times 10^{14}) \cdot (0.1)}{(3.5 \, \text{g/cm}^3)(6 \times 10^{23} \, \text{mol}^{-1}) \cdot (1 \, \text{cm}^2)} \approx 0.7 \, \mu\text{m}. A 1 \, \mu\text{m} sacrificial coating provides a safety margin. Erosion is thus manageable and does not violate fundamental material limits. For denser interstellar patches (n \sim 10^3 \, \text{cm}^{-3} in molecular clouds), the thickness requirement rises to \sim 10 \, \mu\text{m}, which remains feasible for a graphene-based or diamond-like coating.

5. DOPPLER LASER COOLING IN THE ACCELERATION CYCLE

Electronic heat death during the acceleration phase is prevented by anti-Stokes cooling. The ground-based acceleration complex emits two frequencies:

1. Primary beam — for kinetic impulse transfer.
2. Auxiliary beam, spectrally tuned slightly below the resonant frequency of the hull's crystal lattice vibrations.

Atoms of the chip, upon heating, collide with photons of the auxiliary beam and re-emit them with higher energy via the Doppler effect. Heat is continuously extracted from the chip and carried away into space as scattered light. Thus, the chip is forcibly maintained at cryogenic temperatures even within the epicenter of the terawatt acceleration beam.

6. RESOLVING THE DIFFRACTION LIMIT: SWARM AND GRAVITATIONAL LENS

A single nanoprobe cannot form a directional communication beam. We propose launching a Swarm of 10,000 identical "Centaur-Chip" units.

6.1. Coherent Phased Array Synthesis via Quantum-Optical Cross-Correlation

To overcome the diffraction limit across a 4.3 light-year baseline, the swarm of N = 10,000 microprobes must synthesize a monolithic aperture. Let the optical signal emitted or scattered by the i-th probe be modeled as a complex analytic signal: S_{i}(t)=A(t-\tau _{i})e^{i(\omega _{0}t+\phi _{i})}+n_{i}(t) where \tau_i(t) represents the geometric propagation delay relative to the swarm's barycenter, \phi _{i} is the local oscillator phase, \omega _{0} is the carrier frequency, and n_i(t) is the fundamental quantum shot noise. Phase and time-delay synchronization between any pair of nodes (i, j) is achieved through the continuous exchange of entangled or pseudo-random photon sequences. The mutual states are evaluated via the Generalized Cross-Correlation with Phase Transform (GCC-PHAT) framework to extract sub-wavelength tracking parameters: C_{ij}(\Delta)=\int _{-\infty }^{\infty }\frac{\mathcal{F}\{S_{i}(t)\}\mathcal{F}^{*}\{S_{j}(t)\}}{|\mathcal{F}\{S_{i}(t)\}\mathcal{F}^{*}\{S

$_{j}(t)\}|}e^{i\omega \Delta }d\omega$ where $\mathcal{F}$ denotes the Fourier transform. The phase transform whitening filter eliminates amplitude fluctuations, isolating the sharp correlation peak corresponding to the precise differential delay $\Delta_{ij} = \tau_i - \tau_j$. Due to the gravitational gradient of the target stellar system (e.g., Proxima Centauri), the microprobes experience a differential kinematic drift $\delta r_{ij}(t)$ and a corresponding differential Doppler shift $\delta f_{ij}(t) = \frac{\omega_0}{c} \frac{d}{dt}(\delta r_{ij})$. To prevent correlation peak smearing, the dynamic coordinate topology is updated in real-time via multi-lateral laser trilateral beacons. The optimal phase alignment vector $\mathbf{\Phi} = \{\phi_1, \dots, \phi_N\}$ is continuously updated by maximizing the joint pseudo-likelihood function: $\mathcal{L}(\tau ,\mathbf{\Phi })=\sum _{i\neq j}\max _{\Delta }\left|C_{ij}(\Delta)\right|$. This distributed optimization bound allows the swarm to maintain a coherent wavefront accuracy of ≤ 100 nm across a 10 km dynamic baseline.

7. MISSION FINALE SCENARIOS

Scenario A: Quantum Light Capture and Return

The microprobe approaches the target star on a close orbit. Its nanorings switch to absorption mode, harvesting stellar radiation to fully charge the optical accumulator. A gravitational slingshot maneuver combined with accumulated photon ejection fires the chip back toward Earth, or the entire stored stellar energy is used to encode images of the planet into a laser beam transmitted home at light speed.

Scenario B: Non-Returnable Landing

The probe uses accumulated energy to form an ionization field. Entering the exoplanet atmosphere, lateral rings maintain a force-field shape for aerodynamic braking. The 50-gram graphene wafer performs a soft landing, remaining as an eternal research beacon on the surface.

8. ENGINEERING CHALLENGES AND OPEN PROBLEMS

The authors explicitly acknowledge the following non-trivial engineering hurdles:

1. Ultra-High Vacuum Cavity Preparation: Pre-charging cleaning to atomic-level sterility is required to prevent field breakdown at 10^{14} V/m.
2. Swarm Synchronization at 4.3 Light-Years: Mutual quantum-optical correlation has been demonstrated in laboratory settings but never at interstellar distances. Even with post-processing (VLBI-style), determining each chip's position to ~ 100 nm accuracy remains an open problem.
3. Forward Edge Erosion: The sacrificial leading edge must survive continuous relativistic particle bombardment during the braking phase. Analysis indicates $\sim 1 \, \mu\text{m}$ of diamond-like carbon is sufficient for typical ISM densities, but this must be experimentally validated.
4. SGL Receiver Deployment: A dedicated mission to 550 AU is a prerequisite for the communication architecture.
5. Bremsstrahlung and Synchrotron Radiation from the Plasma Wedge: The plasma wedge is created by counter-directed coherent radiation that ablates interstellar gas ahead of the

probe. However, at 0.5c, electrons accelerated in the laser field can emit hard bremsstrahlung and synchrotron radiation. For field intensities approaching $10^{14} \, \text{V/m}$, emitted photons may reach TeV energies. Such gamma-rays have interaction cross-sections orders of magnitude larger than optical photons. The proposed $\sim 1 \, \mu\text{m}$ sacrificial carbon layer is effectively transparent to TeV photons, and the chip electronics would receive a fatal radiation dose within seconds of braking. Mitigation strategies: Perform active deceleration only in regions of extremely low interstellar medium density ($n < 10^{-3} \, \text{cm}^{-3}$), such as the Local Bubble or interarm voids. Rely primarily on gravitational braking (close stellar flyby) instead of photonic thrust, using the accumulated energy only for data transmission. Accept a pure flyby swarm scenario: no braking, but sequential imaging by 10,000 chips over hours, yielding hundreds of frames rather than one. The plasma wedge concept remains valid for dust mitigation but is not a solution for continuous gas braking at 0.5c in typical ISM conditions. These challenges are severe but fall within the domain of engineering — and in some cases, fundamental physics — requiring further study before a flight mission can be designed.

9. CONCLUSION

The proposed "Cyclone-M" class architecture demonstrates that a mission to the exoplanet Proxima Centauri b need not be a "shot into the void." The "Centaur-Chip" becomes not merely a passive sail, but an autonomous relativistic accumulator using stellar light for propulsion, cooling, and communication. The energy density analysis confirms the accumulator operates within classical electrodynamic limits. The sacrificial layer erosion estimate confirms material survivability for direct particle impact. The critical problem of bremsstrahlung radiation from the plasma wedge is identified, and mitigation strategies — including low-density braking corridors, gravitational maneuvering, and flyby-only swarm profiles — are proposed. The concept is ready for detailed numerical modeling and the fabrication of first-generation topological accumulator prototypes in laboratory conditions.

KEYWORDS: Interstellar flight, Lightsail, Optical accumulator, Möbius topology, Doppler cooling, Swarm phased array, Solar Gravity Lens, Schwinger limit, Sacrificial layer erosion, Bremsstrahlung radiation.